

Unsupervised learning: clustering and PCA

K-means and its objective, starting and stopping, choosing K , failure cases, hierarchical clustering, PCA.

THE QUESTION

With no labels at all, how do we find groups in the data, and how do we describe each point with fewer numbers?

THE INTUITION

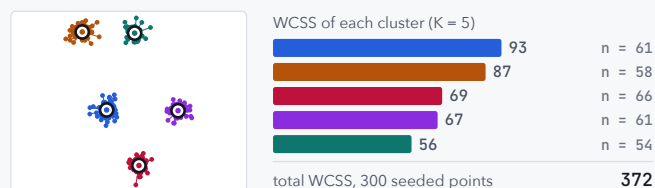
K-means gives each point to the nearest of K centres, moves each centre to the mean of its points, and repeats; both steps lower the total squared distance. PCA keeps the few directions in which the data varies most.

KEY TAKEAWAY

K-means stops at a local minimum of WCSS set by its start: scale features, restart with K-means++, pick K by elbow or silhouette. PCA keeps the covariance's top eigenvectors.

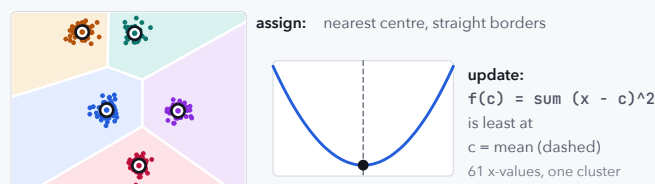
data X: $n \times d$, no labels → standardise → PCA (optional) → K-means++ start → assign & update → WCSS vs K → clusters

1 K-means makes clusters tight



Clusters $C_1, \dots, C_K$ split the n points, each point in exactly one. K-means minimises the within-cluster sum of squares (scikit-learn: inertia) $WCSS = \sum_k \sum_{i \in C_k} \|x_i - \mu_k\|^2$, with μ_k the mean of C_k . Summing over pairs gives the same up to a factor: $\sum_{a,b \in C} \|x_a - x_b\|^2 = 2|C| \sum_{a \in C} \|x_a - \mu\|^2$. Example: $(0,0), (1,0), (0,1)$ have mean $(\frac{1}{3}, \frac{1}{3})$ and WCSS $\frac{4}{3}$.

2 Each step solves half the problem



Fixed centres: WCSS is least when every point joins its nearest centre, which gives Voronoi cells with straight borders. Fixed clusters: the gradient of $\sum_i \|x_i - c\|^2$ is zero at $c = \text{the mean}$. Break ties consistently: $(2,3)$ is at squared distance 5 from both $(1,1)$ and $(4,4)$.

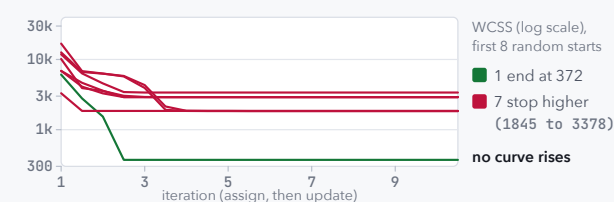
3 Lloyd's algorithm on six points

start: $c_1 = x_1, c_2 = x_4$					after one update ($c = \text{cluster means}$)				
pt	x	$d^2(c_1)$	$d^2(c_2)$	→	$d^2(c_1)$	$d^2(c_2)$	→		
x1	(1, 1)	0	25	C1	0.8125	18.5	C1		
x2	(2, 1)	1	18	C1	0.8125	12.5	C1		
x3	(4, 3)	13	2	C2	7.8125	0.5	C2		
x4	(5, 4)	25	0	C2	17.3125	0.5	C2		
x5	(1, 3)	4	17	C1	1.8125	12.5	C1		
x6	(2, 2)	2	13	C1	0.3125	8.5	C1		

WCSS 9 after assigning, 4.75 after the update labels unchanged: stop

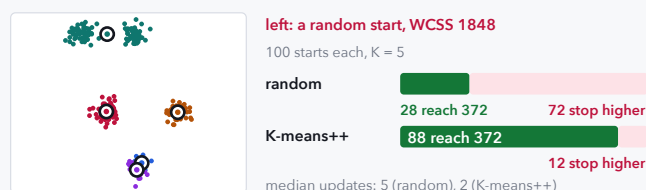
Start from K centres (here x_1 and x_4 ; or from random labels), then assign and update until no label changes. One update moves the centres to $(1.5, 1.75)$ and $(4.5, 3.5)$; after that nothing changes. An iteration costs $O(nKd)$.

4 Neither step can raise WCSS



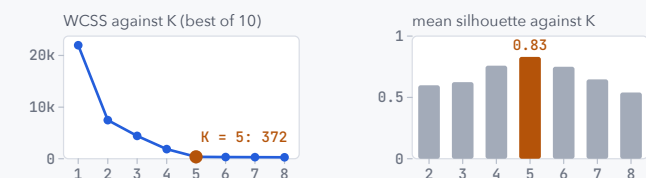
Running example: 300 points, 5 seeded blobs. WCSS is at least 0 and there are at most K^n labellings, so the loop stops, at a local minimum that need not be the best. The lecture's nine points settle after one update (WCSS 16.4 to 0.73).

5 The start decides which minimum you reach



The usual trap: two centres share a blob while one sits between two. K-means++ draws the first centre uniformly, then each next one with probability $D(x)^2 / \sum D^2$, D the distance to the nearest centre so far. Its expected WCSS is within $O(\log K)$ of the best (Arthur and Vassilvitskii, 2007). Keep the best of several starts.

6 Pick K at the bend or by silhouette

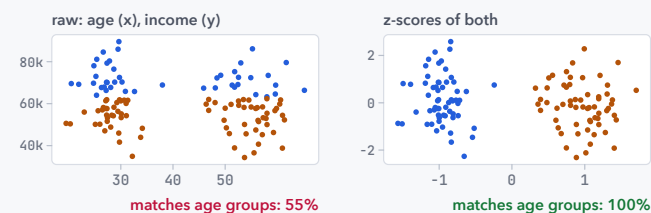


WCSS falls as K grows (0 at $K = n$), so it cannot pick K alone. Ours bends at $K = 5$ (1,870 to 372, then flat); a "largest drop" rule would say $K = 2$, so the elbow is a heuristic. Silhouette: $s(i) = (b - a) / \max(a, b)$, a the mean distance to the own cluster, b to the nearest other; it lies in $[-1, 1]$ and peaks at $K = 5$.

Where K-means fails, other methods, and PCA

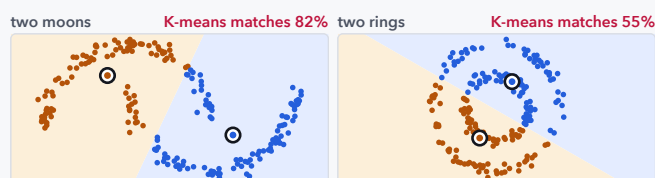
Feature scaling, cluster shapes, hierarchical clustering and DBSCAN, principal components, explained variance.

7 Scale features before measuring distance



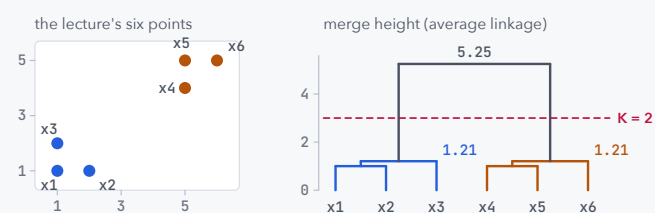
Illustrative data: two age groups (about 28 and 55), income unrelated to age. Income spreads over thousands and age over years, so income decides every distance and K-means splits on it (55% agreement, near chance). On z-scores, $(x - \text{mean}) / \text{sd}$, it finds the age groups exactly. Min-max scaling to $[0, 1]$ also works.

8 Straight borders miss curved clusters



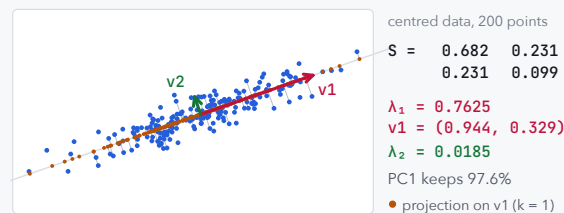
K-means cells are convex, and it favours round clusters of similar size and density. On two moons it matches 82% of points, on two rings 55%. Squared distances also let outliers drag centres. Alternatives: DBSCAN (any shape, marks noise), spectral clustering, Gaussian mixtures (ellipses, soft labels), K-medoids (centres are data points; robust to outliers).

9 Hierarchical clustering records every merge



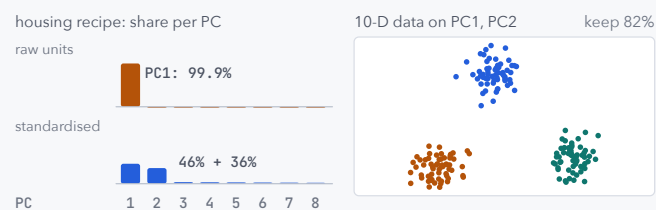
Agglomerative clustering starts from n one-point clusters and repeatedly merges the two closest. Linkage defines "closest": single (nearest pair), complete (farthest pair), average (all pairs), Ward (smallest rise in WCSS). Cutting the tree gives K ; it costs $O(n^2 \log n)$ time and $O(n^2)$ memory. DBSCAN calls a point core if at least MinPts points lie within ϵ and grows clusters through core points; the lecture's ten points ($\epsilon = 1.5$, MinPts 3) give two clusters and one noise point.

10 PCA keeps the highest-variance directions



Centre $\tilde{X} = X - \bar{x}$, form $S = \tilde{X}^T \tilde{X} / (n - 1)$, solve $Sv_i = \lambda_i v_i$ and sort by λ . The scores are $Z = \tilde{X}W_k$ ($n \times k$). PC1 has the largest variance of any direction, PC2 the largest left once orthogonal to PC1. Rebuilding $\hat{x} = W_k z + \bar{x}$ loses $\sum_{i>k} \lambda_i$ (here $\lambda_2 = 0.0185$), the least any k -dimensional linear fit can lose. The PCA notebook's linear autoencoder, trained by gradient descent, lands within 0.6° of v_1 .

11 Units decide the explained variance



The SVD $\tilde{X} = U\Sigma V^T$ gives the same answer more stably: columns of V are the v_i , $\lambda_i = \sigma_i^2 / (n - 1)$, $Z = U_k \Sigma_k$. Keep k for 90 to 95% cumulative share, at the scree-plot elbow, or with $\lambda > 1$ on standardised data. In raw units income alone takes 99.9%; standardised, two PCs keep 82% (size and location). Right: 10-D data; K-means on two PCs gives the same clusters as on all ten.

12 Pause and think

WCSS at $K = n$? 0: each point is its own centre.

Two runs, two answers? Different local minima; keep the lower WCSS.

PCA on z-scores?

Eigenvectors of the correlation matrix.

K-means++ from $(0, 0)$: odds of $(10, 0)$ over $(1, 0)$? 100 : 1, since $D^2 = 100$ vs 1.

Can an update raise WCSS? No: the mean is the best centre.

Why standardise?

Otherwise the widest feature decides.

PC1's share, 2-D example?

$0.7625 / 0.7810 = 97.6\%$.

Silhouette near 0? The point sits on the border of two clusters.

WCSS = $\sum \|x - \mu\|^2$ · assign ↯ update · K-means++ and restarts · elbow or silhouette for K · PCA: centre, eigenvectors of S , keep the top k

Before: the convolutional networks sheet; next: the K-nearest neighbours sheet. Source: lecture slides "Unsupervised learning", "Unsupervised learning: clustering" and "Principal component analysis", and the PCA notebooks (nipunbatra.github.io/ml-teaching). The blobs, age and income, moons, rings and 10-D data are ours (seeded NumPy). Corrected: the six-point example ends at WCSS 4.75 (slides: 4.6, from a table that mixes distances); the three-point WCSS is 4/3 (not 2/3); average linkage joins $\{x_4, x_5\}$ and x_6 at 1.21 (not 1.27); DBSCAN points A, B, C, F have 5, 6, 5, 6 neighbours (clusters unchanged).